

Investigación
de Operaciones

Branch & Bound

Solving integer programming problems

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Integer Programming

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Methodology

Branch & Bound algorithm

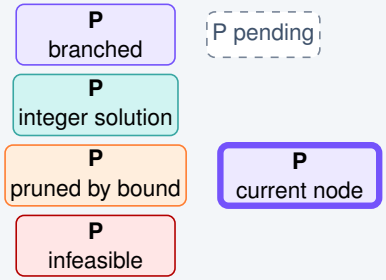
For each subproblem

1. Solve its LP relaxation with **LINGO**.
2. **Fathom** the node if the relaxation is infeasible, if its solution is integer (update the incumbent z^*), or if it does not improve z^*.
3. Si no se poda, **branch** on the first fractional variable $x_j = v$:
   $x_j \leq \lfloor v \rfloor$ y $x_j \geq \lceil v \rceil$.

Conventions

Depth-first search (the $\leq$ branch is explored first).
Subproblems are numbered in order of creation and the *step* gives the order of resolution.

Tree legend



Problem 1

Problem 1 · Statement

$$\begin{aligned} \max\ & z = 5x_1 + 2x_2 \\ \text{s.t.}\ & 3x_1 + x_2 \le 12 \\ & x_1 + x_2 \le 5 \\ & x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 1 · LINGO: root LP (P1)

LINGO model

```
MAX = 5*x1 + 2*x2;
3*x1 + x2 ≤ 12;
x1 + x2 ≤ 5;
END
```

Result

$z = \frac{51}{2} \approx 20.5, \quad x_1 = \frac{7}{2} \approx 3.5,$
$x_2 = \frac{3}{2} \approx 1.5$

Decision

$x_1 = \frac{7}{2}$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 3$ and **P3** with $x_1 \geq 4$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              20.50000
Infeasibilities:                              0.000000
Total solver iterations:                             2
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   3
Nonlinear constraints:               0

Total nonzeros:                      6
Nonlinear nonzeros:                  0

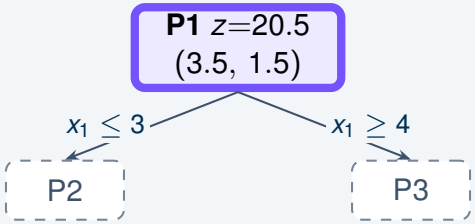
                     Variable           Value        Reduced Cost
                           X1        3.500000            0.000000
                           X2        1.500000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        20.50000            1.000000
                            2        0.000000            1.500000
                            3        0.000000           0.5000000
```

Problem 1 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 1 · Subproblem P2: LINGO

LINGO model

```
MAX = 5*x1 + 2*x2;
3*x1 + x2 ≤ 12;
x1 + x2 ≤ 5;
x1 ≤ 3;
END
```

Result

$z = 19, \quad x_1 = 3, \quad x_2 = 2$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 19$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              19.00000
Infeasibilities:                              0.000000
Total solver iterations:                             1
Time to best solution seconds:                    0.06
Elapsed runtime seconds:                          0.06

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   4
Nonlinear constraints:               0

Total nonzeros:                      7
Nonlinear nonzeros:                  0

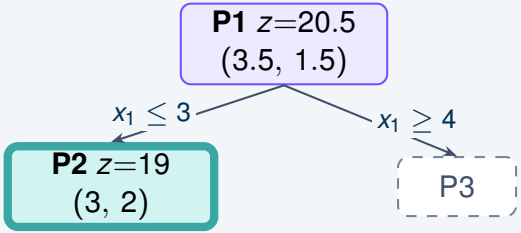

                     Variable           Value        Reduced Cost
                           X1        3.000000            0.000000
                           X2        2.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        19.00000            1.000000
                            2        1.000000            0.000000
                            3        0.000000            2.000000
                            4        0.000000            3.000000
```

Problem 1 · Tree after solving P2

Step 2: P2 solved (integer).

Current incumbent: $z^* = 19$ at $(3, 2)$



Problem 1 · Subproblem P3: LINGO

LINGO model

```
MAX = 5*x1 + 2*x2;
3*x1 + x2 ≤ 12;
x1 + x2 ≤ 5;
x1 ≥ 4;
END
```

Result

$z = 20, \quad x_1 = 4, \quad x_2 = 0$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 20$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              20.00000
Infeasibilities:                              0.000000
Total solver iterations:                             0
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   4
Nonlinear constraints:               0

Total nonzeros:                      7
Nonlinear nonzeros:                  0

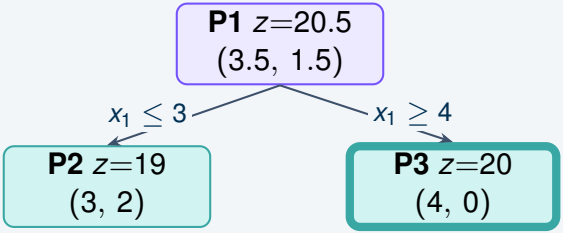

                   Variable           Value        Reduced Cost
                         X1        4.000000            0.000000
                         X2        0.000000            0.000000

                        Row    Slack or Surplus      Dual Price
                          1        20.00000            1.000000
                          2        0.000000            2.000000
                          3        1.000000            0.000000
                          4        0.000000           -1.000000
```

Problem 1 · Tree after solving P3

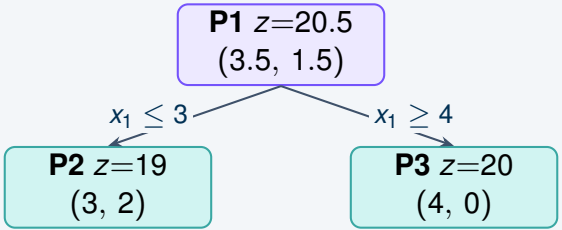
Step 3: P3 solved (integer).

Current incumbent: $z^* = 20$ at $(4, 0)$



Problem 1 · Final solution

Optimal solution: $z^* = 20$, $x_1^* = 4$, $x_2^* = 0$ (3 subproblems solved)



Problem 3

Problem 3 · Statement

$$
\begin{aligned}
&\max\ z = 2x_1 + 3x_2 \\
\text{s.t.}\ \ &x_1 + 2x_2 \le 10 \\
&3x_1 + 4x_2 \le 25 \\
&x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer}
\end{aligned}
$$

Problem 3 · LINGO: root LP (P1)

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 <= 10;
3*x1 + 4*x2 <= 25;
END
```

Result

$z = \frac{35}{2} \approx 17.5, \quad x_1 = 5, \quad x_2 = \frac{5}{2} \approx 2.5$

Decision

$x_2 = \frac{5}{2}$ is fractional $\Rightarrow$ branch: **P2** with $x_2 \leq 2$ and **P3** with $x_2 \geq 3$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              17.50000
Infeasibilities:                              0.000000
Total solver iterations:                             2
Time to best solution seconds:                    0.06
Elapsed runtime seconds:                          0.06

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   3
Nonlinear constraints:               0

Total nonzeros:                      6
Nonlinear nonzeros:                  0

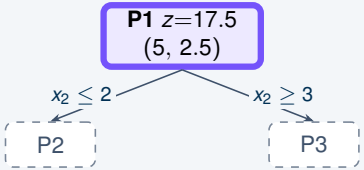

                     Variable           Value        Reduced Cost
                           X1        5.000000            0.000000
                           X2        2.500000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        17.50000            1.000000
                            2        0.000000            0.5000000
                            3        0.000000            0.5000000
```

Problem 3 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 3 · Subproblem P2: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x2 ≤ 2;
END
```

Result

$z = \frac{52}{3} \approx 17.33, \quad x_1 = \frac{17}{3} \approx 5.66, \quad x_2 = 2$

Decision

$x_1 = \frac{17}{3}$ is fractional $\Rightarrow$ branch: **P4** with $x_1 \leq 5$ and **P5** with $x_1 \geq 6$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              17.33333
Infeasibilities:                              0.000000
Total solver iterations:                             1
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   4
Nonlinear constraints:               0

Total nonzeros:                      7
Nonlinear nonzeros:                  0

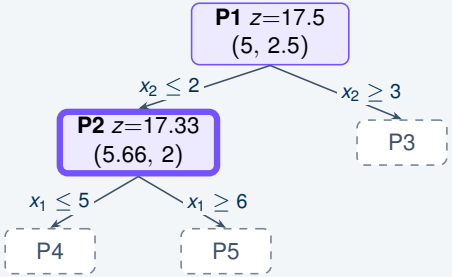

                     Variable           Value        Reduced Cost
                           X1        5.666667            0.000000
                           X2        2.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        17.33333            1.000000
                            2        0.3333333           0.000000
                            3        0.000000            0.6666667
                            4        0.000000            0.3333333
```

Problem 3 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 3 · Subproblem P4: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x1 ≤ 5;
x2 ≤ 2;
END
```

Result

$z = 16, \quad x_1 = 5, \quad x_2 = 2$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 16$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              16.00000
Infeasibilities:                              0.000000
Total solver iterations:                             0
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      8
Nonlinear nonzeros:                  0

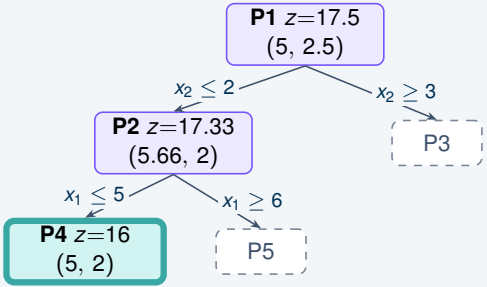

                     Variable           Value        Reduced Cost
                           X1        5.000000            0.000000
                           X2        2.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        16.00000            1.000000
                            2        1.000000            0.000000
                            3        2.000000            0.000000
                            4        0.000000            2.000000
                            5        0.000000            3.000000
```

Problem 3 · Tree after solving P4

Step 3: P4 solved (pruned).

Current incumbent: $z^* = 16$ at $(5, 2)$



Problem 3 · Subproblem P5: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x1 ≥ 6;
x2 ≤ 2;
END
```

Result

$z = \frac{70}{4} \approx 17.25, \quad x_1 = 6, \quad x_2 = \frac{7}{2}1.75$

Decision

$x_2 = \frac{7}{4}$ is fractional $\Rightarrow$ branch: **P6** with $x_2 \leq 1$ and **P7** with $x_2 \geq 2$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              17.25000
Infeasibilities:                              0.000000
Total solver iterations:                             2
Time to best solution seconds:                    0.06
Elapsed runtime seconds:                          0.06

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      8
Nonlinear nonzeros:                  0

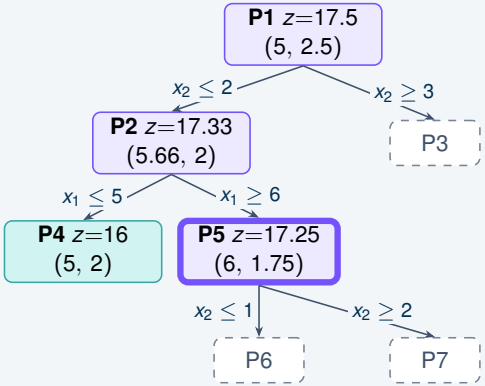

                     Variable           Value        Reduced Cost
                           X1        6.000000            0.000000
                           X2        1.750000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        17.25000            1.000000
                            2       0.5000000            0.000000
                            3       0.000000            0.7500000
                            4       0.000000           -0.2500000
                            5       0.2500000            0.000000
```

Problem 3 · Tree after solving P5

Step 4: P5 solved (branched).

Current incumbent: $z^* = 16$ at $(5, 2)$



Problem 3 · Subproblem P6: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x1 ≥ 6;
x2 ≤ 1;
END
```

Result

$z = 17, \quad x_1 = 7, \quad x_2 = 1$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 17$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              17.00000
Infeasibilities:                              0.000000
Total solver iterations:                             1
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   4
Nonlinear constraints:               0

Total nonzeros:                      7
Nonlinear nonzeros:                  0

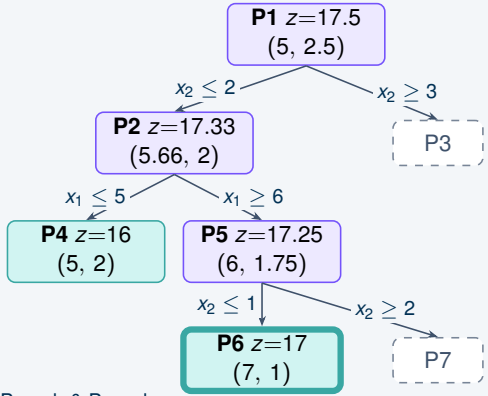

                   Variable           Value        Reduced Cost
                         X1        7.000000            0.000000
                         X2        1.000000            0.000000

                        Row    Slack or Surplus      Dual Price
                          1        17.00000            1.000000
                          2        1.000000            0.000000
                          3        0.000000            0.6666667
                          4        0.000000            0.3333333
```

Problem 3 · Tree after solving P6

Step 5: P6 solved (integer).

Current incumbent: $z^* = 17$ at $(7, 1)$



Problem 3 · Subproblem P7: LINGO

LINGO model

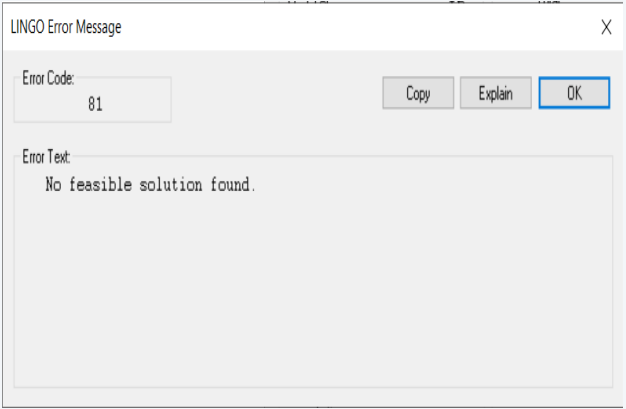
```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x1 ≥ 6;
x2 ≥ 2;
END
```

Result

LINGO reports: **Infeasible**

Decision

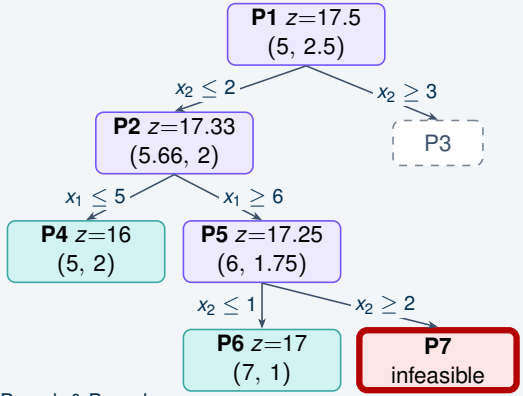
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 3 · Tree after solving P7

Step 6: P7 solved (pruned).

Current incumbent: $z^* = 17$ at $(7, 1)$



Problem 3 · Subproblem P3: LINGO

LINGO model

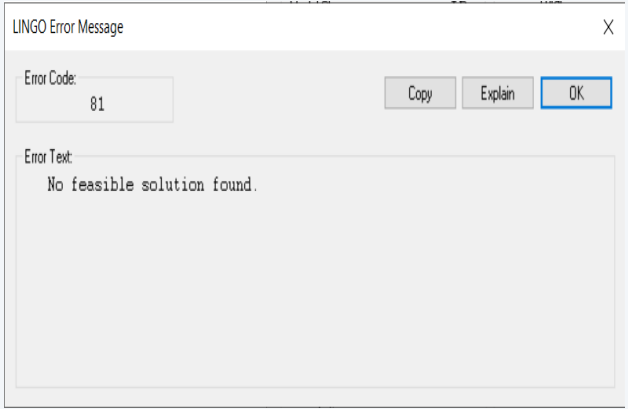
```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x2 ≥ 3;
END
```

Result

$z = 17, \quad x_1 = 4, \quad x_2 = 3$

Decision

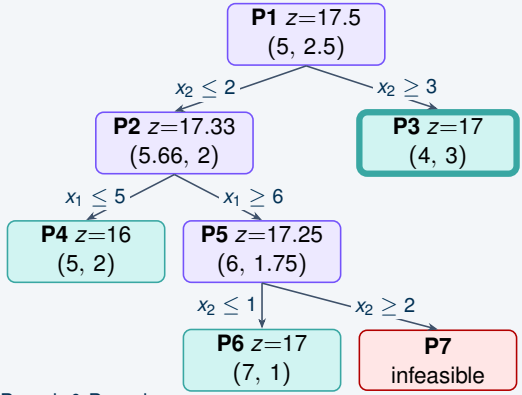
Feasible integer solution, but the current incumbent solution has the same Z value.



Problem 3 · Tree after solving P3

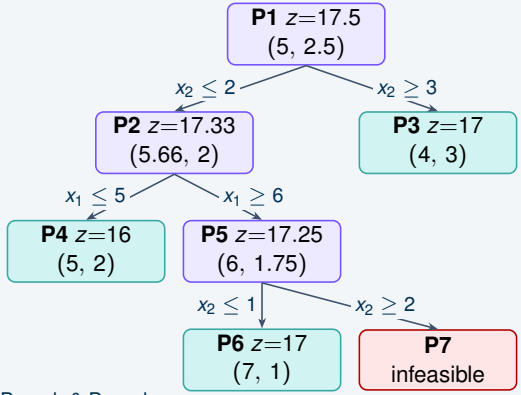
Step 6: P3 solved (integer).

Current incumbent: $z^* = 17$ at $(7, 1)$



Problem 3 · Final solution

Optimal solution: $z^* = 17$, $x_1^* = 7$, $x_2^* = 1$ (7 subproblems solved)



Problem 4

Problem 4 · Statement

$$\max\ z = 4x_1 + 3x_2$$

$$\text{s.t.}\quad 4x_1 + 9x_2 \le 26$$

$$8x_1 + 5x_2 \le 17$$

$$x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer}$$

Problem 4 · LINGO: root LP (P1)

LINGO model

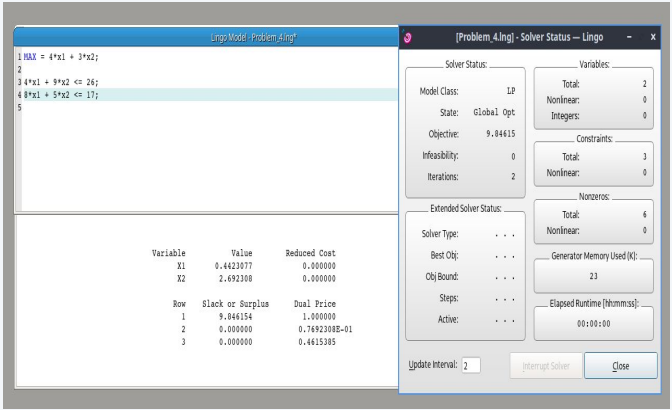
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 <= 26;
8*x1 + 5*x2 <= 17;
END
```

Result

$z = \frac{128}{13} \approx 9.8562, \quad x_1 = \frac{23}{52} \approx 0.4423,$
$x_2 = \frac{35}{13} \approx 2.6923$

Decision

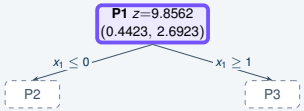
$x_1 = \frac{23}{52}$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 0$ and **P3** with $x_1 \geq 1$.



Problem 4 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 4 · Subproblem P2: LINGO

LINGO model

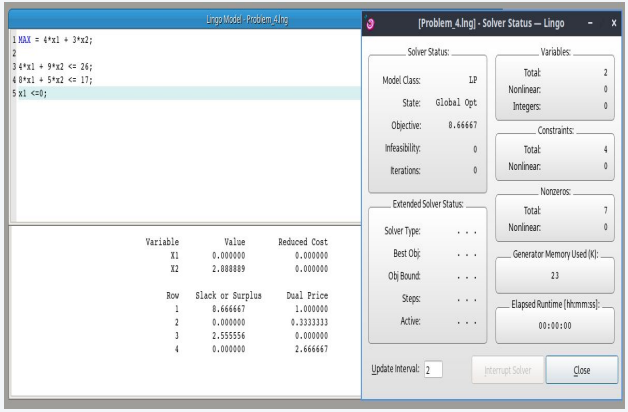
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≤ 0;
END
```

Result

$z = \frac{26}{3} \approx 8.6667, \quad x_1 = 0, \quad x_2 = \frac{26}{9} \approx 2.8889$

Decision

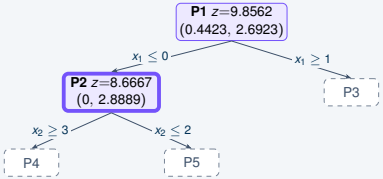
$x_2 = \frac{26}{9}$ is fractional $\Rightarrow$ branch: **P4** with $x_2 \geq 3$ and **P5** with $x_2 \leq 2$.



Problem 4 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 4 · Subproblem P4: LINGO

LINGO model

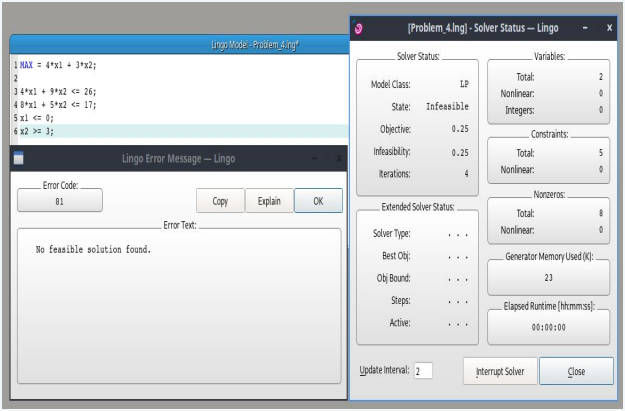
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≤ 0;
x2 ≥ 3;
END
```

Result

LINGO reports: **Infeasible**

Decision

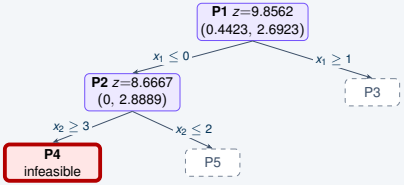
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 4 · Tree after solving P4

Step 3: P4 solved (infeasible).

Current incumbent: none



Problem 4 · Subproblem P5: LINGO

LINGO model

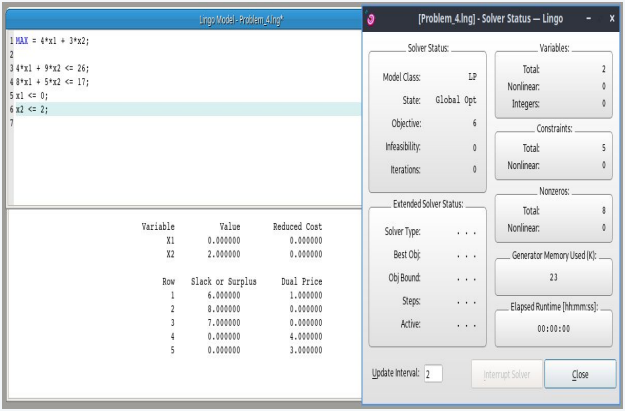
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≤ 0;
x2 ≤ 2;
END
```

Result

$z = 6, \quad x_1 = 0, \quad x_2 = 2$

Decision

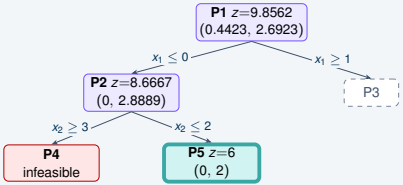
Feasible integer solution ⇒ new **incumbent** $z^* = 6$. Node fathomed.



Problem 4 · Tree after solving P5

Step 4: P5 solved (integer).

Current incumbent: $z^* = 6$ at $(0, 2)$



Problem 4 · Subproblem P3: LINGO

LINGO model

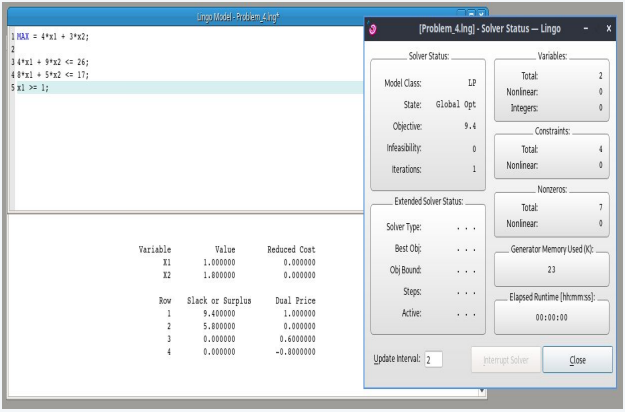
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 <= 26;
8*x1 + 5*x2 <= 17;
x1 >= 1;
END
```

Result

$z = \frac{47}{5} = 9.4, \quad x_1 = 1, \quad x_2 = \frac{9}{5} = 1.8$

Decision

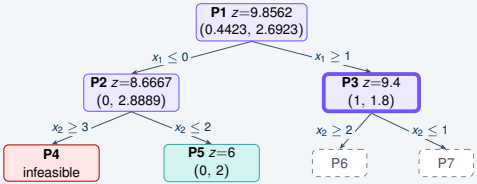
$x_2 = \frac{9}{5}$ is fractional $\Rightarrow$ branch: **P6** with $x_2 \geq 2$ and **P7** with $x_2 \leq 1$.



Problem 4 · Tree after solving P3

Step 5: P3 solved (branched).

Current incumbent: $z^* = 6$ at $(0, 2)$



Problem 4 · Subproblem P6: LINGO

LINGO model

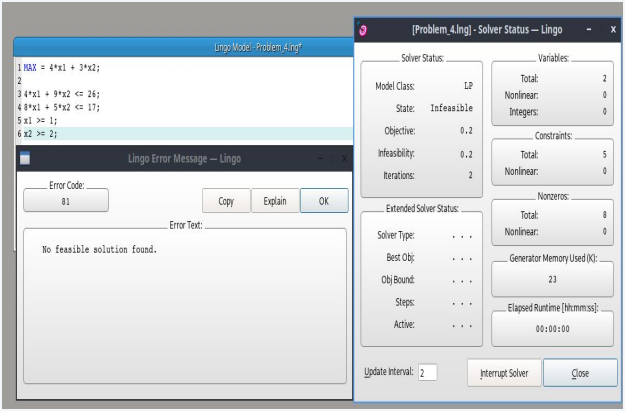
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≥ 1;
x2 ≥ 2;
END
```

Result

LINGO reports: **Infeasible**

Decision

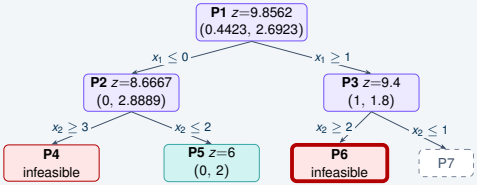
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 4 · Tree after solving P6

Step 6: P6 solved (infeasible).

Current incumbent: $z^* = 6$ at $(0, 2)$



Problem 4 · Subproblem P7: LINGO

LINGO model

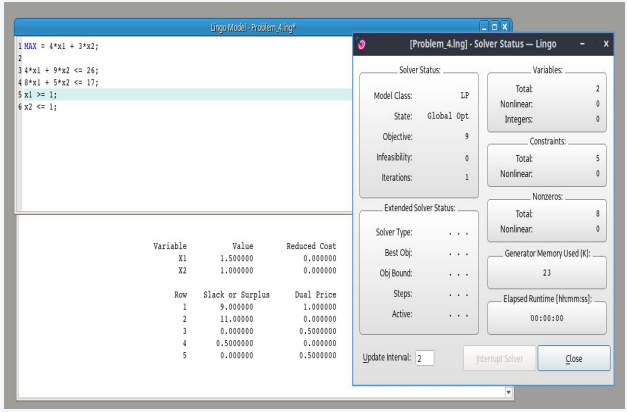
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 <= 26;
8*x1 + 5*x2 <= 17;
x1 >= 1;
x2 <= 1;
END
```

Result

$z = 9, \quad x_1 = \frac{3}{2} = 1.5, \quad x_2 = 1$

Decision

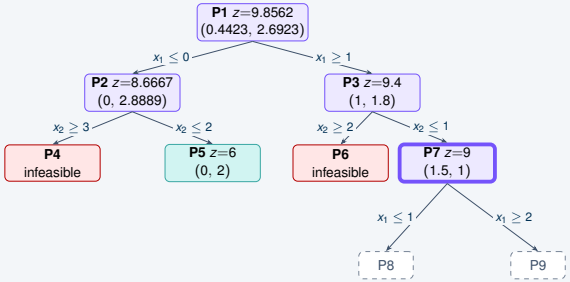
$x_1 = \frac{3}{2}$ is fractional $\Rightarrow$ branch: **P8** with $x_1 \leq 1$ and **P9** with $x_1 \geq 2$.



Problem 4 · Tree after solving P7

Step 7: P7 solved (branched).

Current incumbent: $z^* = 6$ at $(0, 2)$



Problem 4 · Subproblem P8: LINGO

LINGO model

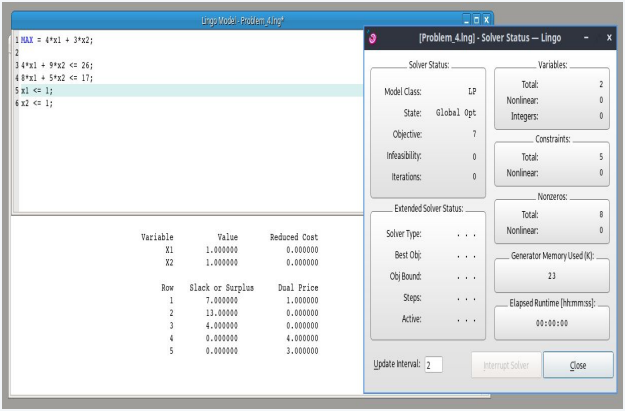
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 <= 26;
8*x1 + 5*x2 <= 17;
x1 <= 1;
x2 <= 1;
END
```

Result

$z = 7, \quad x_1 = 1, \quad x_2 = 1$

Decision

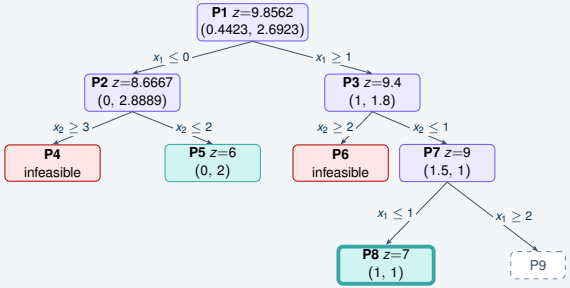
Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 7$. Node fathomed.



Problem 4 · Tree after solving P8

Step 8: P8 solved (integer).

Current incumbent: $z^* = 7$ at $(1, 1)$



Problem 4 · Subproblem P9: LINGO

LINGO model

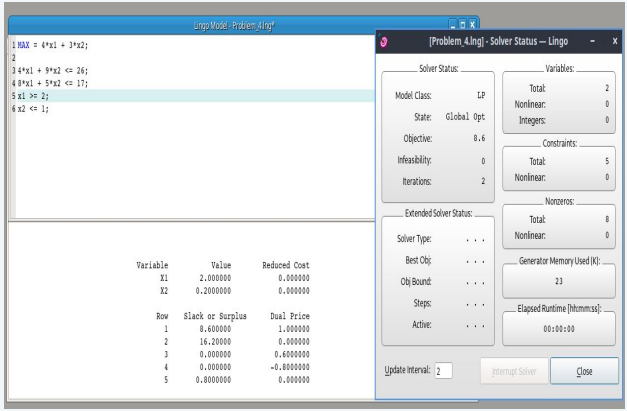
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≥ 2;
x2 ≤ 1;
END
```

Result

$z = \frac{43}{5} = 8.6, \quad x_1 = 2, \quad x_2 = \frac{1}{5} = 0.2$

Decision

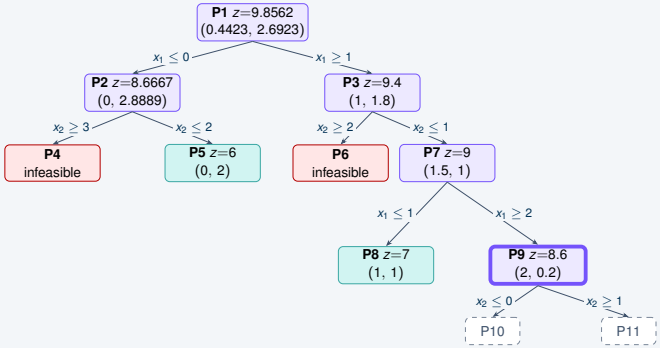
$x_2 = \frac{1}{5}$ is fractional $\Rightarrow$ branch: **P10** with $x_2 \leq 0$ and **P11** with $x_2 \geq 1$.



Problem 4 · Tree after solving P9

Step 9: P9 solved (branched).

Current incumbent: $z^* = 7$ at $(1, 1)$



Problem 4 · Subproblem P10: LINGO

LINGO model

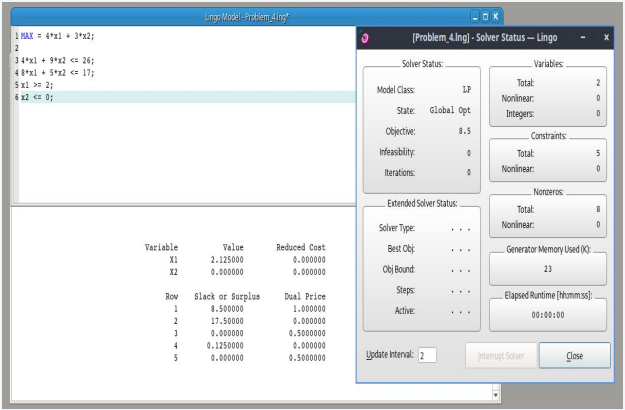
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 <= 26;
8*x1 + 5*x2 <= 17;
x1 >= 2;
x2 <= 0;
END
```

Result

$z = \frac{17}{2} = 8.5, \quad x_1 = \frac{17}{8} = 2.125, \quad x_2 = 0$

Decision

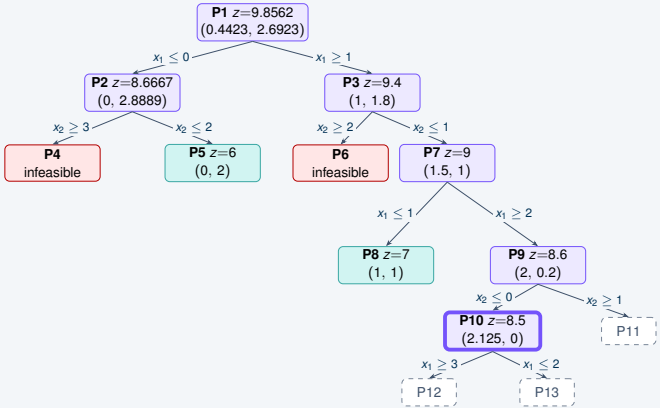
$x_1 = \frac{17}{8}$ is fractional $\Rightarrow$ branch: **P12** with $x_1 \geq 3$ and **P13** with $x_1 \leq 2$.



Problem 4 · Tree after solving P10

Step 10: P10 solved (branched).

Current incumbent: $z^* = 7$ at $(1, 1)$



Problem 4 · Subproblem P12: LINGO

LINGO model

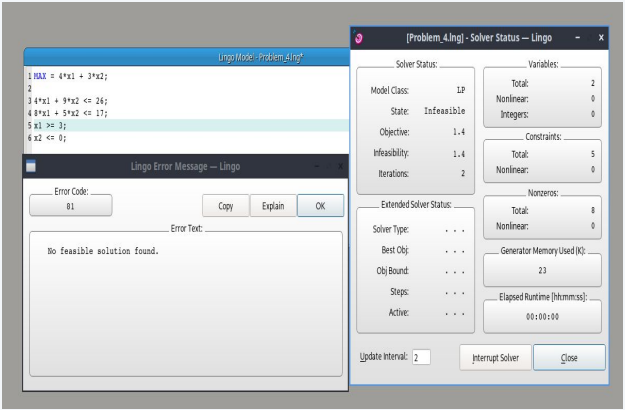
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≥ 3;
x2 ≤ 0;
END
```

Result

LINGO reports: **Infeasible**

Decision

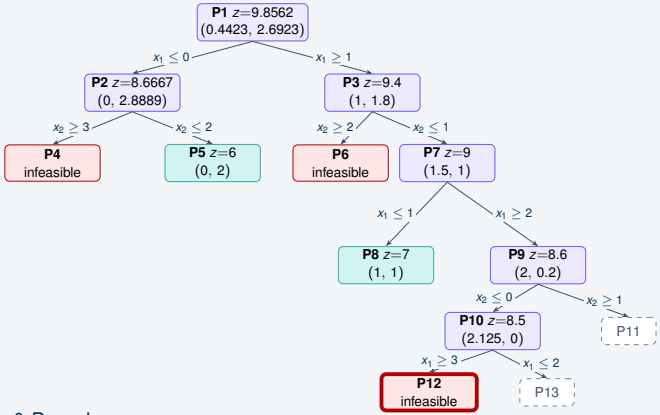
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 4 · Tree after solving P12

Step 11: P12 solved (infeasible).

Current incumbent: $z^* = 7$ at $(1, 1)$



Problem 4 · Subproblem P13: LINGO

LINGO model

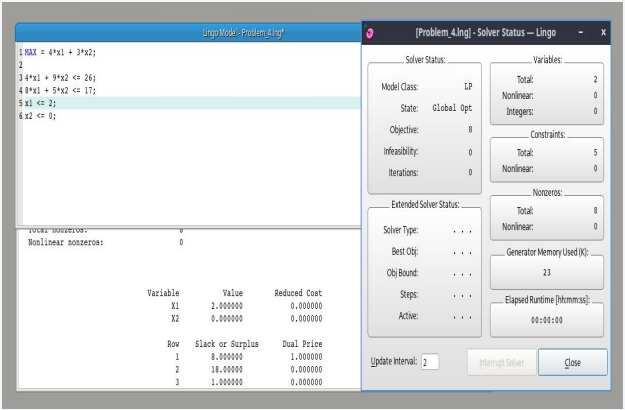
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 <= 26;
8*x1 + 5*x2 <= 17;
x1 <= 2;
x2 <= 0;
END
```

Result

$z = 8, \quad x_1 = 2, \quad x_2 = 0$

Decision

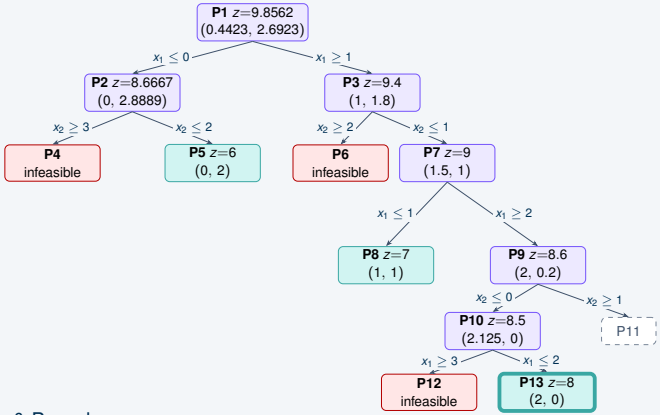
Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 8$. Node fathomed.



Problem 4 · Tree after solving P13

Step 12: P13 solved (integer).

Current incumbent: $z^* = 8$ at $(2, 0)$



Problem 4 · Subproblem P11: LINGO

LINGO model

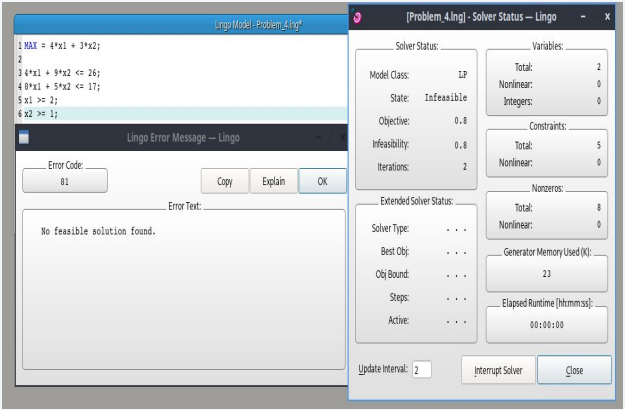
```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
x1 ≥ 2;
x2 ≥ 1;
END
```

Result

LINGO reports: **Infeasible**

Decision

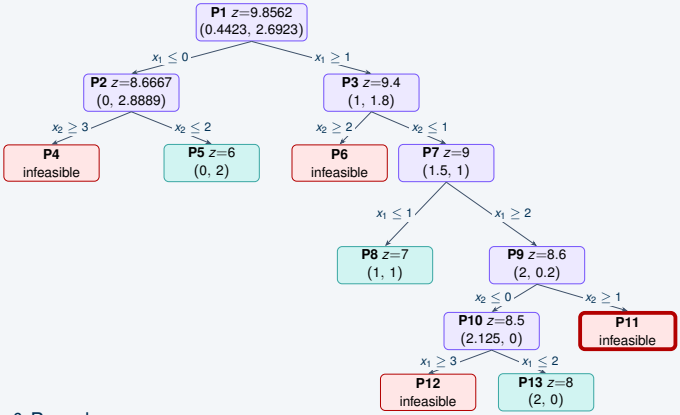
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 4 · Tree after solving P11

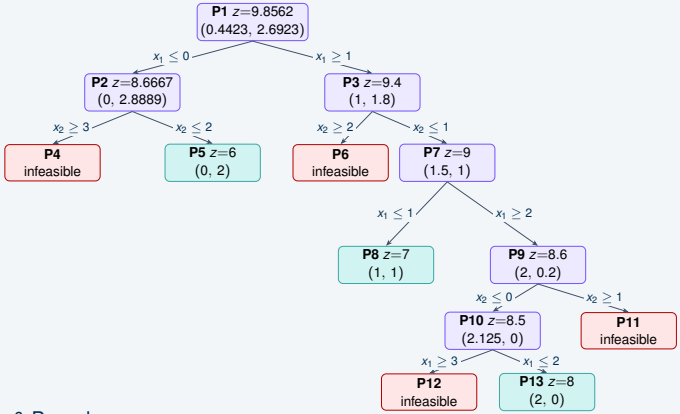
Step 13: P11 solved (infeasible).

Current incumbent: $z^* = 8$ at $(2, 0)$



Problem 4 · Final solution

Optimal solution: $z^* = 8$, $x_1^* = 2$, $x_2^* = 0$ (13 subproblems solved)



Problem 5

Problem 5 · Statement

$$\begin{aligned} & \max\ z = 4x_1 + 5x_2 \\ \text{s.t.}\ & x_1 + 4x_2 \geq 5 \\ & 3x_1 + 2x_2 \geq 7 \\ & x_1, x_2 \geq 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 5 · LINGO: root LP (P1)

LINGO model

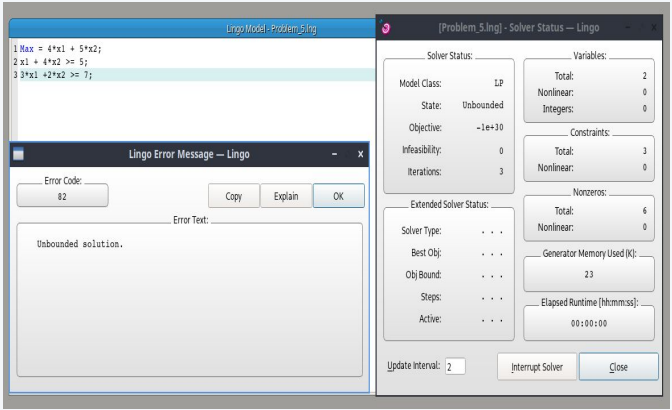
```
MAX = 4*x1 + 5*x2;
x1 + 4*x2 ≥ 5;
3*x1 + 2*x2 ≥ 7;
END
```

Result

LINGO reports: **Unbounded**

Decision

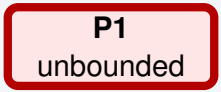
Not Bounded region ⇒ node fathomed by **Unbounded**.



Problem 5 · Tree after solving P1

Step 1: P1 solved (unbounded).

Current incumbent: none



Problem 6

Problem 6 · Statement

$$\begin{aligned} \max\ & z = 4x_1 + 5x_2 \\ \text{s.t.}\ & 3x_1 + 2x_2 \le 10 \\ & x_1 + 4x_2 \le 11 \\ & 3x_1 + 3x_2 \le 13 \\ & x_1, x_2 \ge 0;\ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 6 · Subproblem P1: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 <= 10;
x1 + 4*x2 <= 11;
3*x1 + 3*x2 <= 13;
END
```

Result

$z = \frac{187}{10} \approx 18.7, \quad x_1 = \frac{9}{5} \approx 1.8,$
$x_2 = \frac{23}{10} \approx 2.3$

Decision

$x_1 = \frac{9}{5}$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 1$ and **P3** with $x_1 \geq 2$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              18.70000
Infeasibilities:                              0.000000
Total solver iterations:                             3
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   4
Nonlinear constraints:               0

Total nonzeros:                      8
Nonlinear nonzeros:                  0

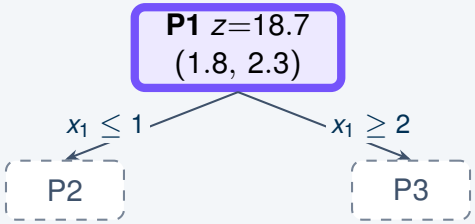

                     Variable           Value        Reduced Cost
                           X1        1.800000            0.000000
                           X2        2.300000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        18.70000            1.000000
                            2        0.000000            1.100000
                            3        0.000000            0.7000000
                            4        0.7000000           0.000000
```

Problem 6 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 6 · Subproblem P2: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 ≤ 10;
x1 + 4*x2 ≤ 11;
3*x1 + 3*x2 ≤ 13;
x1 ≤ 1;
END
```

Result

$z = \frac{33}{2} \approx 16.5, \quad x_1 = 1, \quad x_2 = \frac{5}{2} \approx 2.5$

Decision

$x_2 = \frac{5}{2}$ is fractional $\Rightarrow$ branch: **P4** with $x_2 \leq 2$ and **P5** with $x_2 \geq 3$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              16.50000
Infeasibilities:                              0.000000
Total solver iterations:                             1
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      9
Nonlinear nonzeros:                  0

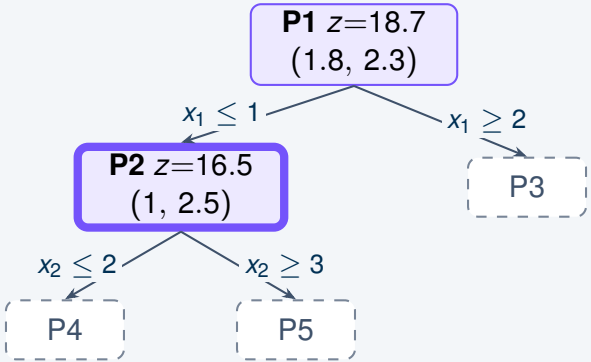

                     Variable           Value        Reduced Cost
                           X1        1.000000            0.000000
                           X2        2.500000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        16.50000            1.000000
                            2        2.000000            0.000000
                            3        0.000000            1.250000
                            4        2.500000            0.000000
                            5        0.000000            2.750000
```

Problem 6 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 6 · Subproblem P4: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 ≤ 10;
x1 + 4*x2 ≤ 11;
3*x1 + 3*x2 ≤ 13;
x1 ≤ 1;
x2 ≤ 2;
END
```

Result

$z = 14, \quad x_1 = 1, \quad x_2 = 2$

Decision

Feasible integer solution ⇒ new **incumbent** $z^* = 14$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              14.00000
Infeasibilities:                              0.000000
Total solver iterations:                             0
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   6
Nonlinear constraints:               0

Total nonzeros:                     10
Nonlinear nonzeros:                  0

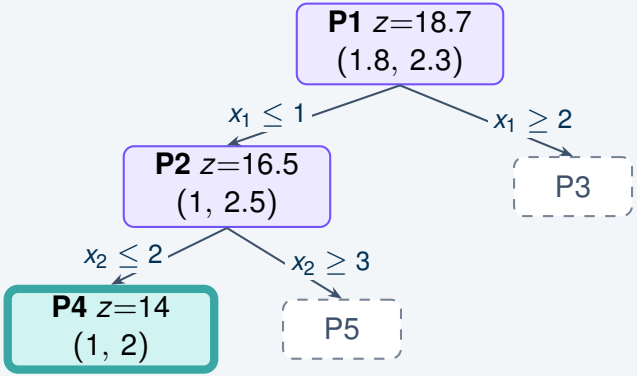

                     Variable           Value        Reduced Cost
                           X1        1.000000            0.000000
                           X2        2.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        14.00000            1.000000
                            2        3.000000            0.000000
                            3        2.000000            0.000000
                            4        4.000000            0.000000
                            5        0.000000            4.000000
                            6        0.000000            5.000000
```

Problem 6 · Tree after solving P4

Step 3: P4 solved (integer).

Current incumbent: $z^* = 14$ at $(1, 2)$



Problem 6 · Subproblem P5: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 ≤ 10;
x1 + 4*x2 ≤ 11;
3*x1 + 3*x2 ≤ 13;
x1 ≤ 1;
x2 ≥ 3;
END
```

Result

LINGO reports: **Infeasible**

Decision

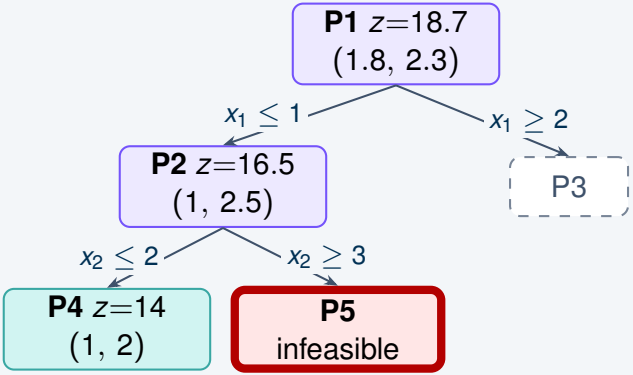
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 6 · Tree after solving P5

Step 4: P5 solved (infeasible).

Current incumbent: $z^* = 14$ at $(1, 2)$



Problem 6 · Subproblem P3: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 ≤ 10;
x1 + 4*x2 ≤ 11;
3*x1 + 3*x2 ≤ 13;
x1 ≥ 2;
END
```

Result

$z = 18, \quad x_1 = 2, \quad x_2 = 2$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 18$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              18.00000
Infeasibilities:                              0.000000
Total solver iterations:                             2
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      9
Nonlinear nonzeros:                  0

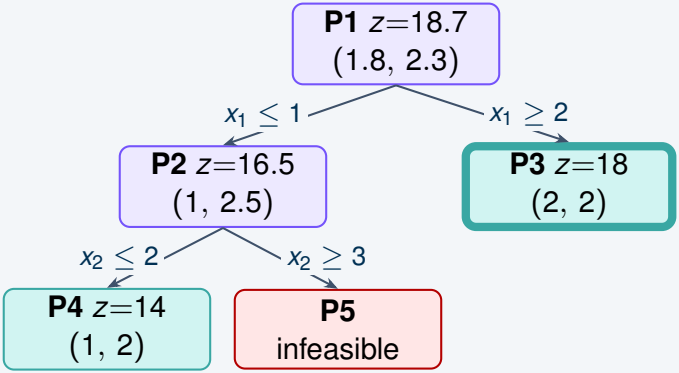

                     Variable           Value        Reduced Cost
                           X1        2.000000            0.000000
                           X2        2.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        18.00000            1.000000
                            2        0.000000            2.500000
                            3        1.000000            0.000000
                            4        1.000000            0.000000
                            5        0.000000           -3.500000
```

Problem 6 · Tree after solving P3

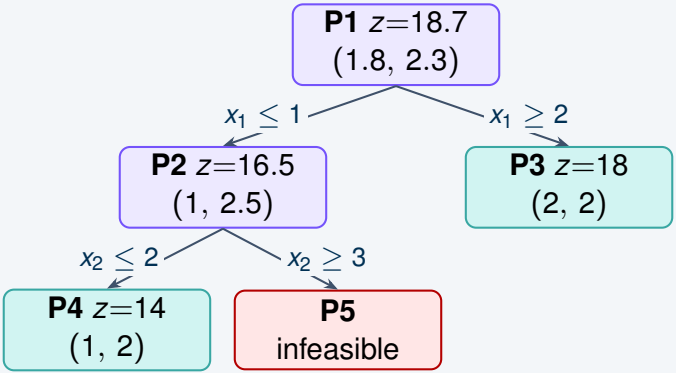
Step 5: P3 solved (integer).

Current incumbent: $z^* = 18$ at $(2, 2)$



Problem 6 · Final solution

Optimal solution: $z^* = 18$, $x_1^* = 2$, $x_2^* = 2$ (5 subproblems solved)



Problem 7

Problem 7 · Statement

$$\begin{aligned} &\max\ z = 7x_1 + 3x_2 \\ \text{s.t.}\ \ &2x_1 + x_2 \le 9 \\ &3x_1 + 2x_2 \le 13 \\ &x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 7 · Subproblem P1: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
END
```

Result

$z = \frac{91}{3} \approx 30.33, \quad x_1 = \frac{13}{3} \approx 4.33, \quad x_2 = 0$

Decision

$x_1 = \frac{13}{3}$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 4$ and **P3** with $x_1 \geq 5$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              30.33333
Infeasibilities:                              0.000000
Total solver iterations:                             2
Time to best solution seconds:                    0.06
Elapsed runtime seconds:                          0.06

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   3
Nonlinear constraints:               0

Total nonzeros:                      6
Nonlinear nonzeros:                  0

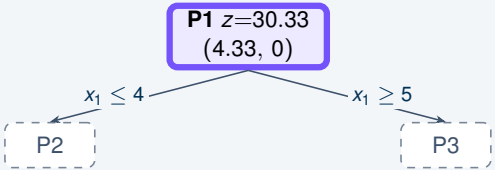

                     Variable           Value        Reduced Cost
                           X1        4.333333            0.000000
                           X2        0.000000            1.666667

                          Row    Slack or Surplus      Dual Price
                            1        30.33333            1.000000
                            2        0.3333333           0.000000
                            3        0.000000            2.333333
```

Problem 7 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 7 · Subproblem P2: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
END
```

Result

$z = \frac{59}{2} \approx 29.5, \quad x_1 = 4, \quad x_2 = \frac{1}{2} \approx 0.5$

Decision

$x_2 = \frac{1}{2}$ is fractional $\Rightarrow$ branch: **P4** with $x_2 \leq 0$ and **P5** with $x_2 \geq 1$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              29.50000
Infeasibilities:                              0.000000
Total solver iterations:                             1
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   4
Nonlinear constraints:               0

Total nonzeros:                      7
Nonlinear nonzeros:                  0

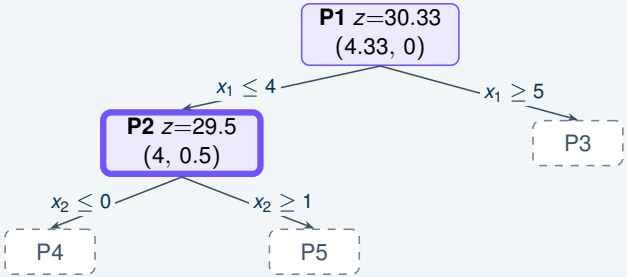

                   Variable           Value        Reduced Cost
                         X1        4.000000            0.000000
                         X2       0.5000000            0.000000

                        Row    Slack or Surplus      Dual Price
                          1        29.50000            1.000000
                          2       0.5000000            0.000000
                          3        0.000000            1.500000
                          4        0.000000            2.500000
```

Problem 7 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 7 · Subproblem P4: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
x2 ≤ 0;
END
```

Result

$z = 28, \quad x_1 = 4, \quad x_2 = 0$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 28$. Node fathomed.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              28.00000
Infeasibilities:                              0.000000
Total solver iterations:                             0
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      8
Nonlinear nonzeros:                  0

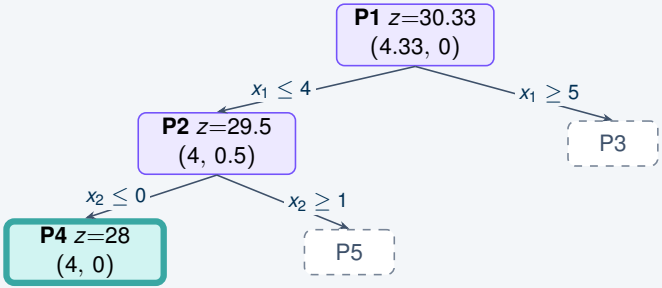

                     Variable           Value        Reduced Cost
                           X1        4.000000            0.000000
                           X2        0.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        28.00000            1.000000
                            2        1.000000            0.000000
                            3        1.000000            0.000000
                            4        0.000000            7.000000
                            5        0.000000            3.000000
```

Problem 7 · Tree after solving P4

Step 3: P4 solved (integer).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P5: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
x2 ≥ 1;
END
```

Result

$z = \frac{86}{3} \approx 28.67, \quad x_1 = \frac{11}{3} \approx 3.67, \quad x_2 = 1$

Decision

$x_1 = \frac{11}{3}$ is fractional $\Rightarrow$ branch: **P6** with $x_1 \leq 3$ and **P7** with $x_1 \geq 4$.

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              28.66667
Infeasibilities:                              0.000000
Total solver iterations:                             2
Time to best solution seconds:                    0.06
Elapsed runtime seconds:                          0.06

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      8
Nonlinear nonzeros:                  0

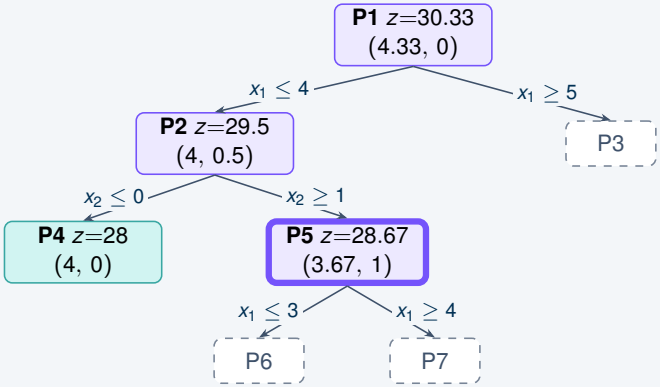
                      Variable           Value        Reduced Cost
                            X1        3.666667            0.000000
                            X2        1.000000            0.000000

                           Row    Slack or Surplus      Dual Price
                             1        28.66667            1.000000
                             2       0.6666667            0.000000
                             3        0.000000            2.333333
                             4       0.3333333            0.000000
                             5        0.000000           -1.666667
```

Problem 7 · Tree after solving P5

Step 4: P5 solved (branched).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P6: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 3;
x2 ≥ 1;
END
```

Result

$z = 27, \quad x_1 = 3, \quad x_2 = 2$

Decision

$z = 27$ does not improve the incumbent $z^* = 28$ ($\leq$) $\Rightarrow$ pruned by **bound** (the solution is integer but not better).

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134

Licensee info: Eval Use Only
License expires:  6 MAY 2027

Global optimal solution found.
Objective value:                              27.00000
Infeasibilities:                              0.000000
Total solver iterations:                             1
Time to best solution seconds:                    0.07
Elapsed runtime seconds:                          0.07

Model Class:                                        LP

Total variables:                     2
Nonlinear variables:                 0
Integer variables:                   0

Total constraints:                   5
Nonlinear constraints:               0

Total nonzeros:                      8
Nonlinear nonzeros:                  0

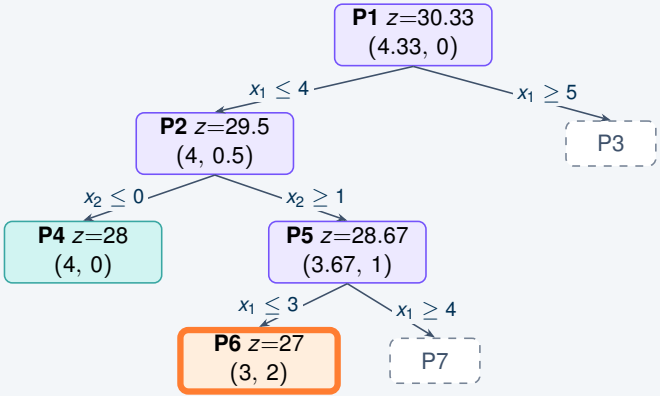

                     Variable           Value        Reduced Cost
                           X1        3.000000            0.000000
                           X2        2.000000            0.000000

                          Row    Slack or Surplus      Dual Price
                            1        27.00000            1.000000
                            2        1.000000            0.000000
                            3        0.000000            1.500000
                            4        0.000000            2.500000
                            5        1.000000            0.000000
```

Problem 7 · Tree after solving P6

Step 5: P6 solved (pruned).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P7: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
x2 ≥ 1;
x1 ≥ 4;
END
```

Result

LINGO reports: **Infeasible**

Decision

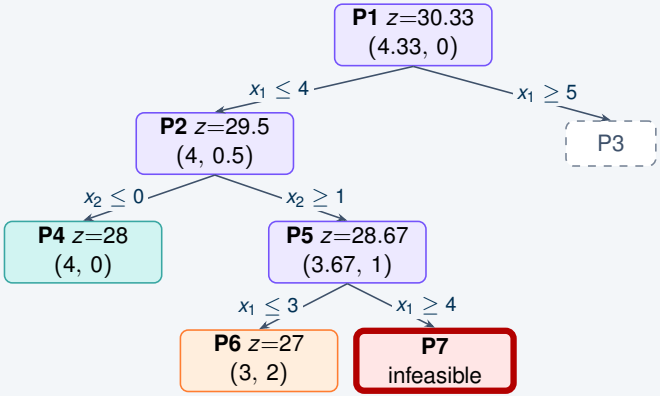
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 7 · Tree after solving P7

Step 6: P7 solved (infeasible).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P3: LINGO

LINGO model

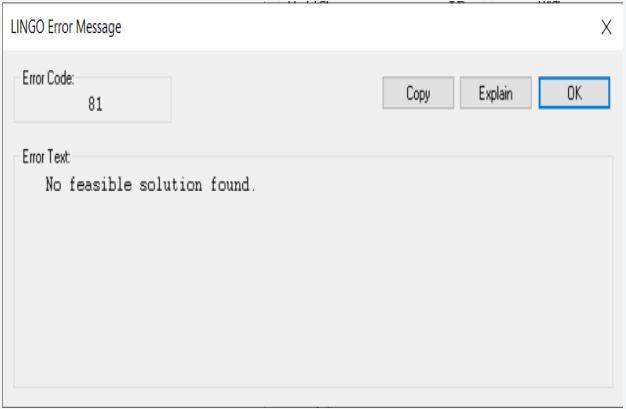
```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≥ 5;
END
```

Result

LINGO reports: **Infeasible**

Decision

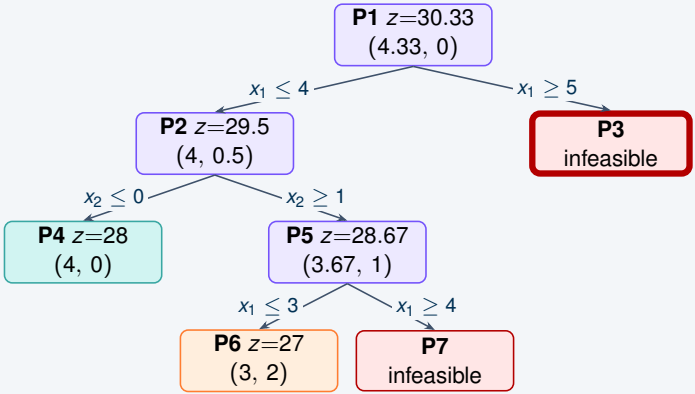
Empty feasible region ⇒ node fathomed by **infeasibility**.



Problem 7 · Tree after solving P3

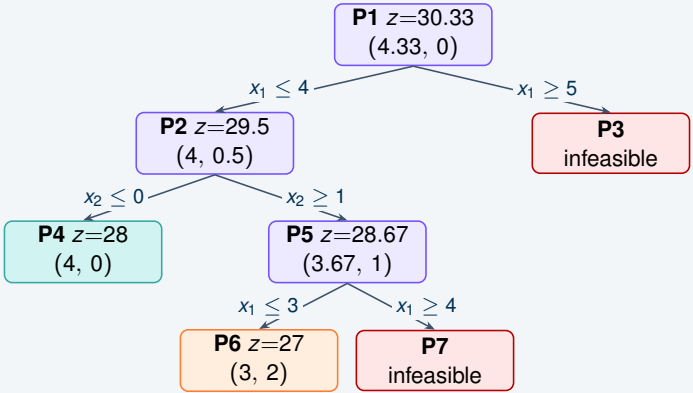
Step 7: P3 solved (infeasible).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Final solution

Optimal solution: $z^* = 28$, $x_1^* = 4$, $x_2^* = 0$ (7 subproblems solved)



Summary

Summary of results

Problem	Objective	Subproblems	z^*	(x_1^*, x_2^*)
Problem 1	max	3	20	(4, 0)
Problem 3	max	7	17	(7, 1)
Problem 4	max	13	8	(2, 0)
Problem 5	max	1	Unbounded	
Problem 6	max	5	18	(2, 2)
Problem 7	max	7	28	(4, 0)



Investigación
de Operaciones

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Integer Programming

November 8, 2026